TANISHA PRITHA

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production RAG pipelines, multi-agent orchestration systems, and LLM-powered backends. Skilled in FastAPI, async Python, and PostgreSQL-based retrieval. Focused on retrieval accuracy, structured output, observability, and cost-efficient inference in real systems.

TECHNICAL SKILLS

AI / GenAI: LLMs, Production RAG, Retrieval Engineering, Agentic Workflows, Structured Output (Pydantic), Prompt Engineering, AI Evaluation (RAGAS), Observability & Tracing (LangSmith), Semantic Caching, Fallback & Retry Strategies
AI Infra: LiteLLM, Cohere Rerank, Pinecone, pgvector, FAISS, ChromaDB, BAAI/bge, all-MiniLM, OpenRouter, Deepgram STT
Backend: Python, FastAPI, Django, REST APIs, Async I/O, JWT, RBAC, Redis, Supabase, Background Tasks, ETL Pipelines
Frontend: React.js, Next.js, TypeScript
Data & Infra: PostgreSQL, Redis, Docker, AWS EC2, Vercel, Streamlit, Scikit-learn, XGBoost, PaddleOCR
Strengths: Retrieval Quality, Cost Optimization, AI System Observability, Security-First Engineering

EXPERIENCE

AI Engineer Intern | AptlyHired (Startup) | Nov 2025 – Dec 2025 | Remote, New Delhi

- Built an end-to-end **RAG pipeline** with multi-format ingestion (PDF, DOCX, TXT, PNG, JPG), PaddleOCR, semantic chunking, pgvector retrieval, and an Upload → Store → Chat workflow in Django.
- Improved retrieval accuracy through optimized chunk sizing, noise filtering, and prompt design; integrated into a web UI used by internal testers.

PROJECTS

Threadbase | AI Content Scheduling Platform | FastAPI, PostgreSQL, Redis, Supabase, Clerk | GitHub | Live App

- Built an **LLM-powered content engine** that generates and schedules social posts from user ideas and media; exposed a REST API (draft, schedule, publish) with Clerk auth and JWT-protected endpoints.
- Designed a **Redis-backed job queue** with FastAPI BackgroundTasks and Redis pub/sub for async scheduling; idempotency keys prevent duplicate jobs and TTL-based cache invalidation reduces redundant LLM calls.
- Tracked job lifecycle (enqueued, executing, delivered) in PostgreSQL for a full audit trail; Supabase handles media storage with signed URL access.

Multi-Agent Company Compliance Engine | FastAPI, pgvector, LangSmith, Cohere Rerank, RAGAS | GitHub | Live App

- Architected a **Planner-Reasoner-Verifier multi-agent system** for forensic auditing of compliance documents; agent consensus across all three agents required before decisions are finalized, cutting hallucination risk in high-stakes outputs.
- Built a **production RAG pipeline** with semantic chunking, BAAI/bge embeddings, pgvector retrieval, and **Cohere Rerank** to cut top-K from 20 to 5; enforced **Pydantic structured output** (risk level, violations, confidence) on all LLM responses.
- Routed LLM calls through **LiteLLM** with provider fallback (OpenRouter to Mistral); instrumented with **LangSmith tracing** (latency, tokens, trace IDs) and a **semantic cache** (cosine sim >= 0.95) to reduce redundant Pinecone queries.
- Measured retrieval quality with **RAGAS** (faithfulness, context precision, answer relevancy) on a 15-question eval set.

Smart Predictive Maintenance System | Python, XGBoost, Scikit-learn, Streamlit | GitHub | Live App

- Built an **end-to-end ML pipeline**: ETL (load, clean, validate), feature engineering (temp_diff, power, wear_rate), model training, and real-time inference on UCI AI4I 2020 industrial sensor data.
- Trained and benchmarked Logistic Regression, Random Forest, and XGBoost for **binary failure classification**; best model achieved **0.98 accuracy and 0.85 F1**; Random Forest Regressor for RUL estimation achieved **MAE 0.00389**.
- Deployed a **Streamlit dashboard** on Hugging Face Spaces for real-time sensor input and failure probability visualization.

EDUCATION

B.Tech in Mechanical Engineering | Government College of Engineering, Nagpur | Graduating: June, 2027

CERTIFICATIONS

Introduction to Machine Learning: Art of the Possible (AWS) | HackerRank SQL (Intermediate) | Data Modeling for Amazon DocumentDB (AWS)